

Semantic Interoperability, Ontology Alignment, and Semantic Fidelity

Why Two Systems Can Exchange Data Without Meaning the Same Thing

Knowledge Systems and Reality Drift - Note #4 v1

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The Basic Pattern

Modern systems are expected to exchange information across organizations, databases, disciplines, and software platforms. Technical interoperability allows data to move. Syntactic interoperability allows the receiving system to parse its structure. Semantic interoperability aims for something harder: preserving meaning across the exchange.

Ontology alignment supports this goal by matching classes, properties, and entities that appear to refer to the same things. One system's "client" may be mapped to another's "customer." A clinical term may be mapped across coding systems. Two graphs may reconcile differently named entities.

But a successful mapping does not establish full equivalence. Terms can overlap while carrying different boundaries, histories, purposes, and consequences. Data can arrive in the correct field and still mean something different from what the receiving system assumes.

When Exchange Is Mistaken for Understanding

Alignment systems often rely on lexical similarity, shared instances, structural position, expert crosswalks, or learned embeddings. These methods can identify useful correspondences without preserving every contextual distinction.

The practical pressure is toward completion. Organizations need a mapping so records can be merged, reports generated, and services coordinated. Ambiguities are therefore resolved into operational equivalences. "Close enough to exchange" becomes "the same."

Once integrated data begins supporting decisions, the mapping acquires authority. Differences that were explicit in separate systems disappear inside the unified representation. Interoperability succeeds while semantic fidelity weakens.

How Alignment Drift Emerges

Stage 1 - Heterogeneity

Communities develop vocabularies and schemas for their own purposes and contexts.

Stage 2 - Mapping

Terms, classes, and relations are connected through crosswalks, matching rules, or learned similarity.

Stage 3 - Operational Equivalence

Mapped elements are treated as interchangeable so data can be queried and combined.

Stage 4 - Context Loss

Differences in scope, intent, time, granularity, and institutional use are no longer carried with the data.

Stage 5 - Drift

The integrated system reports coherent results whose apparent uniformity was created by the mapping itself.

Semantic Fidelity and Ontology Alignment

Semantic Fidelity is the degree to which meaning survives transformation. In ontology alignment, fidelity has several dimensions. Lexical fidelity concerns words. Referential fidelity concerns whether terms point to the same entities. Causal fidelity concerns whether relationships preserve the same mechanisms. Pragmatic fidelity concerns whether a term performs the same function in each institutional setting.

A mapping may succeed lexically while failing pragmatically. It may identify the same person but misread the status attached to that person. It may preserve a diagnosis label while losing the evidentiary threshold under which it was assigned.

The central safeguard is not perfect equivalence. It is visible non-equivalence. Mappings should preserve confidence, direction, provenance, version, exceptions, and the dimensions along which meaning was compressed.

The Representation Stack

Local experience becomes a local term. The term becomes a coded field. The field is mapped into a shared ontology. The aligned data becomes a cross-system metric or model input. Each step increases interoperability while potentially removing the context needed to interpret the original claim.

By the top of the stack, agreement may reflect a common mapping rather than a common reality. The systems converge because differences have been normalized out of the representation.

Examples Across Systems

Healthcare: Two diagnostic codes may be mapped as equivalents even though they reflect different criteria, levels of certainty, or billing practices.

Government: Agencies may exchange categories such as household, employment, or disability while defining them differently for distinct programs.

Business Systems: “Active customer” may mean recent purchase, paid subscription, account login, or sales qualification depending on the source.

Artificial Intelligence: Graph alignment can merge relations from heterogeneous sources and present the result as a single coherent knowledge base.

Conclusion

Semantic interoperability is not achieved merely because information moves cleanly between systems. The deeper question is whether the distinctions that made the information meaningful survived the movement.

Two systems can exchange data, share identifiers, and answer common queries while remaining semantically misaligned. Interoperability becomes reliable only when the mapping remains answerable to the contexts it compresses.

Related Phrases and Concepts

- semantic interoperability
- ontology alignment
- ontology matching
- schema matching

- semantic mapping
- data integration
- vocabulary crosswalk
- semantic equivalence

Semantic Fidelity Project Resources

- [Research Library \(GitHub\)](#)
- [Articles & Essays \(Substack\)](#)
- [Semantic Fidelity Glossary](#)